24V$_{OUT}$ High Current Boost WLED Driver

Features

- 2.3V to 24V Input Voltage
- Drives WLED strings up to 24V
- Internal 4A Switch Current Limit
- 1.2MHz Fixed Switching Frequency
- Internal soft-start limits the inrush current
- Dimming frequency for CTRL pin: 20kHz~1MHz
- Output Short Circuit Protection
- Output Over-Voltage Protection
- Thermal Shutdown Protection
- -40°C to +85°C Temperature Range
- Available in SOT23-6 Package

Applications

- GPS Navigation Systems
- Tablet Personal Computer
- LCD Bias Supply
- Handheld Devices
- Portable Media Player (PMP)
- General Purposes

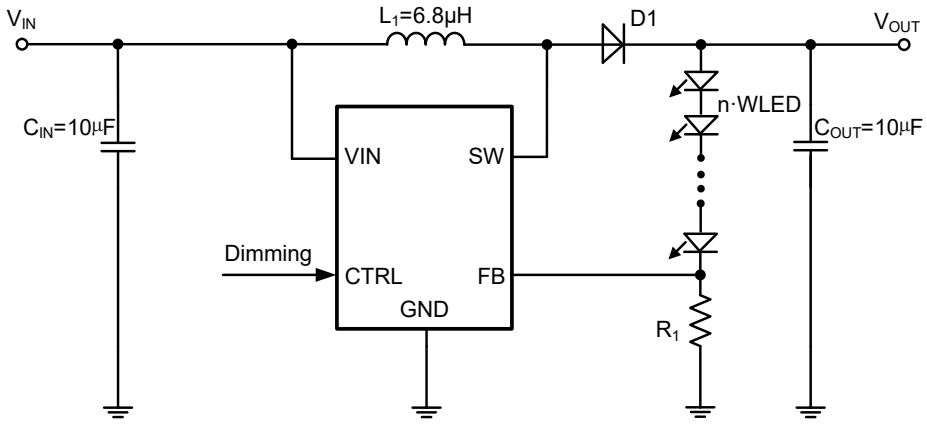
General Description

The RY3725 is DC/DC step-up converter that deliver an accurate constant current for driving LEDs. Operation at a fixed switching frequency of 1.2MHz allows the device to be used with small value external ceramic capacitors and inductor. LEDs connected in series are driven with a regulated current set by the external resistor.

The RY3725 includes under-voltage lockout, current limiting, and thermal overload protection to prevent damage in the event of an output overload. The RY3725 is ideal for driving WLED series up to 24V.

The RY3725 is available in a small 6-pin SOT-23 package.

Typical Application Circuit

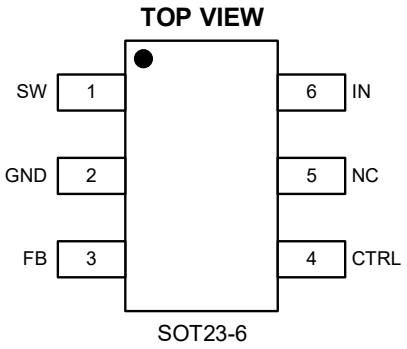


Typical Application Circuit

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Package and Pin Description

Pin Configuration



Top Marking: KKYLL (device code: KK, Y=year code, LL= lot number code)

Pin Description

Pin	Name	Function
1	SW	Power Switch Output. SW is the drain of the internal MOSFET switch. Connect the power inductor and output rectifier to SW.
2	GND	Ground Pin
3	FB	Feedback Reference Voltage Pin. Series connect a resistor between WLED and ground as a current sense. Sense the current feedback voltage to set the current rating.
4	CTRL	Enable and dimming control. When used as enable input, pull high to turn on IC. When used as dimming input, the first pulse should be longer than 200ns to turn on IC. And the recommend dimming frequency range is 20kHz~1MHz.
5	NC	Not Connected.
6	IN	Input Supply Pin

Order Information (1)

Marking	Part No.	Model	Description	Package	T/R Qty
KKYLL	70390024	RY3725BT6	RY3725BT6 WLED Driver, V_{IN} 2.3-24V, V_{OUT} V_{IN}-24V, 1MHz, VFB 0.2V, SOT23-6	SOT23-6	3000PCS

Note (1): All RYCHIP parts are Pb-Free and adhere to the RoHS directive.

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Specifications

Absolute Maximum Ratings (1) (2)

Item	Min	Max	Unit
V_{IN} CTRL voltage	-0.3	30	V
SW voltage	-0.3	31	V
All Other Pins	–0.3	6	V
Power dissipation (3) @T_A=25°C	0.6		W
Operating junction temperature, T_J	-40	150	°C
Storage temperature, T_{stg}	–65	150	°C
Lead Temperature (Soldering, 10sec.)		260	°C

Note (1): Exceeding these ratings may damage the device.

Note (2): The device is not guaranteed to function outside of its operating conditions.

Note (3): The maximum allowable power dissipation is a function of the maximum junction temperature, $T_{J(MAX)}$, the junction-to-ambient thermal resistance, $R_{\theta JA}$, and the ambient temperature, T_A. The maximum allowable power dissipation at any ambient temperature is calculated using: $P_{D\,(MAX)} = (T_{J(MAX)} - T_A)/R_{\theta JA}$. Exceeding the maximum allowable power dissipation causes excessive die temperature, and the regulator goes into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage. Thermal shutdown engages at T_J=160°C (typical) and disengages at T_J= 130°C (typical).

ESD Ratings

Item	Description	Value	Unit
$V_{(ESD-HBM)}$	Human Body Model (HBM) ANSI/ESDA/JEDEC JS-001-2014 Classification, Class: 2	±2000	V
$V_{(ESD-CDM)}$	Charged Device Mode (CDM) ANSI/ESDA/JEDEC JS-002-2014 Classification, Class: C0b	±200	V
$I_{LATCH-UP}$	JEDEC STANDARD NO.78E APRIL 2016 Temperature Classification, Class: I	±150	mA

Recommended Operating Conditions

Item	Min	Max	Unit
Operating junction temperature (1)	–40	125	°C
Operating temperature range	-40	85	°C
Input voltage V_{IN}	2.3	24	V
Output voltage V_{OUT}		24	V

Note (1): All limits specified at room temperature (TA = 25°C) unless otherwise specified. All room temperature limits are 100% production tested. All limits at temperature extremes are ensured through correlation using standard Statistical Quality Control (SQC) methods. All limits are used to calculate Average Outgoing Quality Level (AOQL).

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Thermal Information

Item	Description	Value	Unit
$R_{\theta JA}$	Junction-to-ambient thermal resistance (1)(2)	105	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	55	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	17.5	°C/W
ψ_{JT}	Junction-to-top characterization parameter	3.5	°C/W
ψ_{JB}	Junction-to-board characterization parameter	17.5	°C/W

Note (1): The package thermal impedance is calculated in accordance to JESD 51-7.

Note (2): Thermal Resistances were simulated on a 4-layer, JEDEC board

Electrical Characteristics (1) (2)

V_{IN}=5V, T_A=25°C, unless otherwise specified.

Parameter	Test Conditions	Min	Typ.	Max	Unit
Input voltage range		2.3		24	V
Output voltage range				24	V
Supply Current (Quiescent)	V_{FB} =110%		150	200	μA
Supply Current (Shutdown)	V_{CTRL} =0 or CTRL = GND		1	10	μA
Feedback Voltage		0.190	0.200	0.210	mV
SW On Resistance			80	150	mΩ
Current Limit	Duty cycle=50%		4		A
Switching Frequency			1.2		MHz
Maximum Duty Cycle	V_{FB}=90%		90		%
Minimum On-Time			100		nS
CTRL Rising Threshold		1.1			V
CTRL Falling Threshold				0.9	V
PWM Dimming Frequency		20		1000	kHz
Under-Voltage Lockout Threshold	Wake up V_{IN} Voltage		2.0	2.1	V
	Shutdown V_{IN} Voltage	1.7	1.8		V
	Hysteresis V_{IN} voltage		200		mV
Thermal Shutdown			160		℃
Thermal Hysteresis			30		℃

Note (1): MOSFET on-resistance specifications are guaranteed by correlation to wafer level measurements.

Note (2): Thermal shutdown specifications are guaranteed by correlation to the design and characteristics analysis.

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Functions Description

Under-Voltage Lockout (UVLO)

Under-voltage lockout (UVLO) protects the chip from operating at an insufficient supply voltage. UVLO protection monitors the internal regulator voltage. When the voltage is lower than UVLO threshold voltage, the device is shut off. When the voltage is higher than UVLO threshold voltage, the device is enabled again.

Enable and PWM Dimming

When the input voltage is above maximal UVLO rising threshold and the CTRL pin is pulled high, the RY3725 is enabled. When the CTRL pin is pulled low, the RY3725 goes into shutdown mode. In shutdown mode, less than 1μA input current is consumed. Because there is a conductive path from the input to the output through the inductor and Schottky diode, the output voltage is equal to the input voltage during shutdown. The CTRL pin allows disabling and enabling of the device as well as brightness control of the LEDs by applying a PWM signal from 10kHz to 1MHz. When a PWM signal is applied, the LED current is turned on when the CTRL is high and off when CTRL is pulled low. Changing the CTRL duty cycle therefore changes the LED brightness.

Soft-Start

The RY3725 begins soft start when the CTRL pin is pulled high. at the beginning of the soft start period, the isolation FET is turned on slowly to charge the output capacitor. After the pre-charge phase, the RY3725 starts switching. This is called switching soft start phase. An internal soft start circuit limits the peak inductor current according to the output voltage. The switching soft start phase is about 1.5ms typically. The soft start function reduces the inrush current during startup.

Thermal Shutdown

Thermal shutdown prevents the chip from operating at exceedingly high temperatures. When the silicon die temperature exceeds 160°C, it shuts down the whole chip. When the temperature falls below its lower threshold (Typ. 130°C) the chip is enabled again.

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LED Applications Information

Setting the LED Current

The LED current is controlled by the feedback resistor, R_1. The current through the LEDs is given by the equation

$$V_{FB}/R1$$

I_{LED} is average LED current. According to the Superposition Theorem, we can implement the analog dimming function, and we provide a variety of different FB voltage to choose from, which can help customers more easily adapt to their application needs.

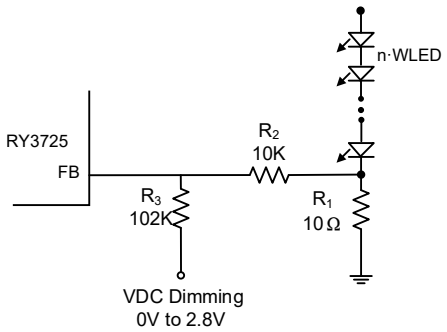
Dimming Control

1. Using a PWM Signal to CTRL Pin

For controlling the LED brightness, the RY3725 can perform the dimming control by applying a PWM signal to CTRL pin. The internal soft start and the wide range dimming frequency can eliminate inrush current and audio noise when dimming. The average LED current is proportional to the PWM signal duty cycle. The magnitude of the PWM signal should be higher than the minimum enables voltage of CTRL pin, in order to let the dimming control perform correctly for preventing the flicker issue, the selected PWM frequency is from 10kHz to 1MHz.

2. Using a DC Voltage

When CTRL remains high level, using a variable DC voltage to adjust the brightness is a popular method in some applications. According to the Superposition Theorem, as the DC voltage increases, the voltage contributed to VFB increases and the voltage drop on R2 decreases, i.e. the LED current decreases. For example, RY3725 can use a DC voltage ranging from 0V-2.8V to adjust the LED current, the selection of resistors sets dimming control of LED current from 20mA to 0mA.



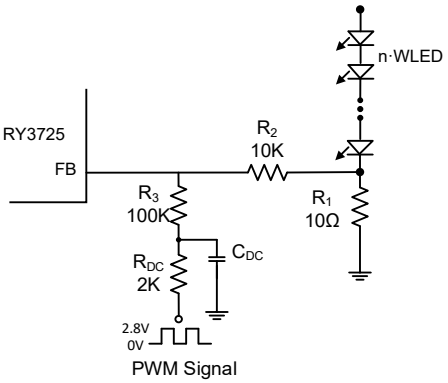
The LED current can be calculated by the following Equation:

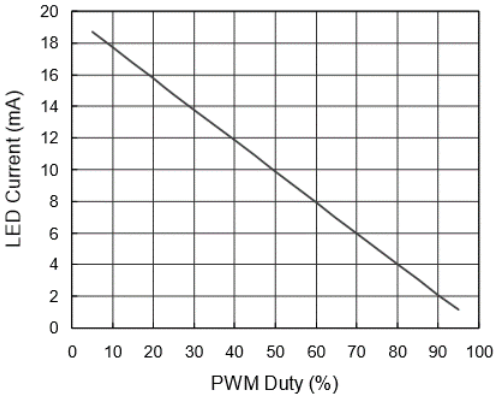
$$I_{LED} = \frac{V_{FB} - \frac{R_2 \times (V_{DC} - V_{FB})}{R_3}}{R_1}$$

3. Using a Filtered PWM signal

Another common application is using a filtered PWM signal as an adjustable DC voltage for LED dimming control. A filtered PWM signal acts as the DC voltage to regulate the output current. In this circuit, the output ripple depends on the frequency of PWM signal. To fix the frequency of PWM signal and change the duty cycle of PWM signal can get different output current.

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The LED current can be calculated by the following Equation:

$$I_{LED} = \frac{V_{FB} - \frac{R_2 \times (V_{PWM} \times Duty - V_{FB})}{R_3 + R_{DC}}}{R_1}$$

Inductor Selection

Because the selection of the inductor affects steady state operation, transient behavior, and loop stability, the inductor is the most important component in power regulator design. There are three important inductor specifications, inductor value, saturation current, and dc resistance (DCR).

The RY3725 is designed to work with inductor values between 2.2μH and 22μH. Follow the Equation below to calculate the inductor's peak current for the application. To calculate the current in the worst case, use the minimum input voltage, maximum output voltage, and maximum load current of the application. To have enough design margin, choose the inductor value with -30% tolerance, and a low power-conversion efficiency for the calculation. In a boost regulator, the inductor dc current can be calculated with Equation:

$$I_{L(DC)} = \frac{V_{OUT} \times I_{OUT}}{V_{IN} \times \eta}$$

Where

- V_{OUT} = output voltage
- I_{OUT} = output current
- V_{IN} = input voltage
- η = power conversion efficiency, use 80% for most applications

The inductor ripple current is calculated with the Equation 3 for an asynchronous boost converter in continuous conduction mode (CCM).

$$\Delta I_{L(P-P)} = \frac{V_{IN} \times (V_{OUT} + V_{DIODE} - V_{IN})}{L \times F_S \times (V_{OUT} + V_{DIODE})}$$

Where

- $\Delta I_{L(P-P)}$ = inductor ripple current
- L = inductor value
- V_{IN} = input voltage
- V_{OUT} = output voltage
- V_{DIODE} = output diode forward voltage
- F_S = switching frequency, Hertz

Therefore, the inductor peak current is calculated with Equation:

$$I_{L(P)} = I_{L(DC)} + \frac{\Delta I_{L(P-P)}}{2}$$

Normally, it is advisable to work with an inductor peak-to-peak current of less than 40% of the average inductor current for maximum output current. A smaller ripple from a larger valued inductor reduces the magnetic hysteresis losses in the inductor and EMI. Bit in the same way, load transient response time is increased.

Because the RY3725 is for relatively small output current application, the inductor peak-to-peak current could be as high as 200% of the average current with a small inductor value, which means the RY3725 always works in DCM mode.

Input and Output Capacitor Selection

The output capacitor is mainly selected to meet the requirements for output ripple and loop stability. This ripple voltage is related to the capacitor's capacitance and its equivalent series resistance (ESR). Assuming a ceramic capacitor with zero ESR, the minimum capacitance needed for a given ripple can be calculated by:

$$C_{OUT} = \frac{I_{OUT} \times D_{MAX}}{F_S \times V_{RIPPLE}}$$

where

- D_{MAX} = maximum switching duty cycle
- I_{RIPPLE} = peak to peak output voltage ripple

The ESR impact on the output ripple must be considered if tantalum or aluminum electrolytic capacitors are used. Care must be taken when evaluating a ceramic capacitor's derating under dc bias, aging, and ac signal. For example, the dc bias can significantly reduce capacitance. A ceramic capacitor can lose more than 50% of its capacitance at its rated voltage. Therefore, always leave margin on the voltage rating to ensure adequate capacitance at the required output voltage.

It is recommended to use the output capacitor with effective capacitance in the range of 10μF to 22μF. The output capacitor affects the small signal control loop stability of the boost regulator. If the output capacitor is below the range, the boost regulator can potentially become unstable. Increasing the output capacitor makes the output voltage ripple smaller in PWM mode.

For input capacitor, a ceramic capacitor with more than 1.0μF is enough for most applications.

Diode Selection

A Schottky diode should be used for the output diode. The forward current rating of the diode should be higher than the load current, and the reverse voltage rating must be higher than the output voltage. Do not use ordinary rectifier diodes, since slow switching speeds and long recovery times cause the efficiency and the load regulation to suffer.

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Layout Guidelines

PC board layout is an important part of DC-DC converter design. Poor board layout can disrupt the performance of a DC-DC converter and surrounding circuitry by contributing to EMI, ground bounce, and resistive voltage loss in the traces. These can send erroneous signals to the DC-DC converter resulting in poor regulation or instability. Good layout can be implemented by following a few simple design rules.

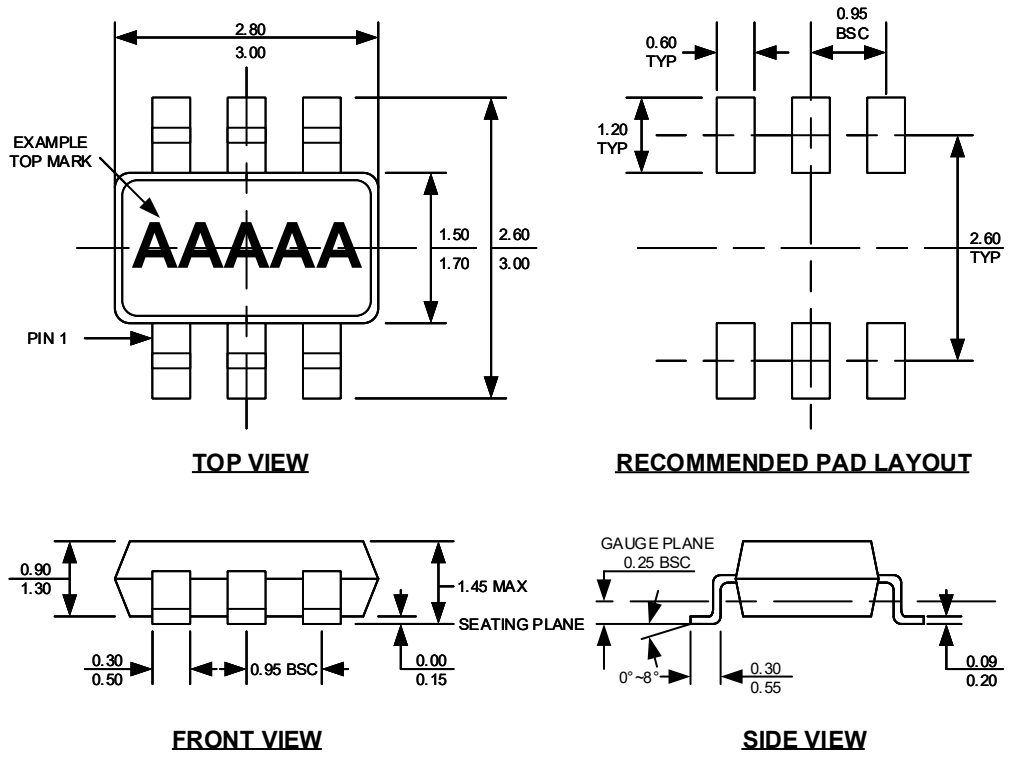
1. Minimize area of switched current loops. In a buck regulator there are two loops where currents are switched rapidly. The first loop starts from the CIN input capacitor, to the regulator VIN terminal, to the regulator SW terminal, to the inductor then out to the output capacitor COUT and load. The second loop starts from the output capacitor ground, to the regulator GND terminals, to the inductor and then out to COUT and the load. To minimize both loop areas the input capacitor should be placed as close as possible to the VIN terminal. Grounding for both the input and output capacitors should consist of a small localized topside plane that connects to GND. The inductor should be placed as close as possible to the SW pin and output capacitor.
2. Minimize the copper area of the switch node. The SW terminals should be directly connected with a trace that runs on top side directly to the inductor. To minimize IR losses this trace should be as short as possible and with an enough width. However, a trace that is wider than 100 mils will increase the copper area and cause too much capacitive loading on the SW terminal. The inductors should be placed as close as possible to the SW terminals to further minimize the copper area of the switch node.
3. Have a single point ground for all device analog grounds. The ground connections for the feedback components should be connected then routed to the GND pin of the device. This prevents any switched or load currents from flowing in the analog ground plane. If not properly handled, poor grounding can result in degraded load regulation or erratic switching behavior.
4. Minimize trace length to the FB terminal. The feedback trace should be routed away from the SW pin and inductor to avoid contaminating the feedback signal with switch noise.
5. Make input and output bus connections as wide as possible. This reduces any voltage drops on the input or output of the converter and can improve efficiency. If voltage accuracy at the load is important make sure feedback voltage sense is made at the load. Doing so will correct for voltage drops at the load and provide the best output accuracy.

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Package Description

SOT23-6



NOTE:
1. CONTROL DIMENSION IS IN INCHES. DIMENSION IN BRACKET IS IN MILLIMETERS.
2. PACKAGE LENGTH DOES NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
3. PACKAGE WIDTH DOES NOT INCLUDE INTERLEAD FLASH OR PROTRUSIONS.
4. LEAD COPLANARITY (BOTTOM OF LEADS AFTER FORMING) SHALL BE 0.004" INCHES MAX.
5. DRAWING CONFORMS TO JEDEC MS-012, VARIATION BA.
6. DRAWING IS NOT TO SCALE.

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Modify Information

Version	Modifications	Date
V1.0.1	Update Electrical Characteristics（Feedback Voltage）	2025.09.24

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